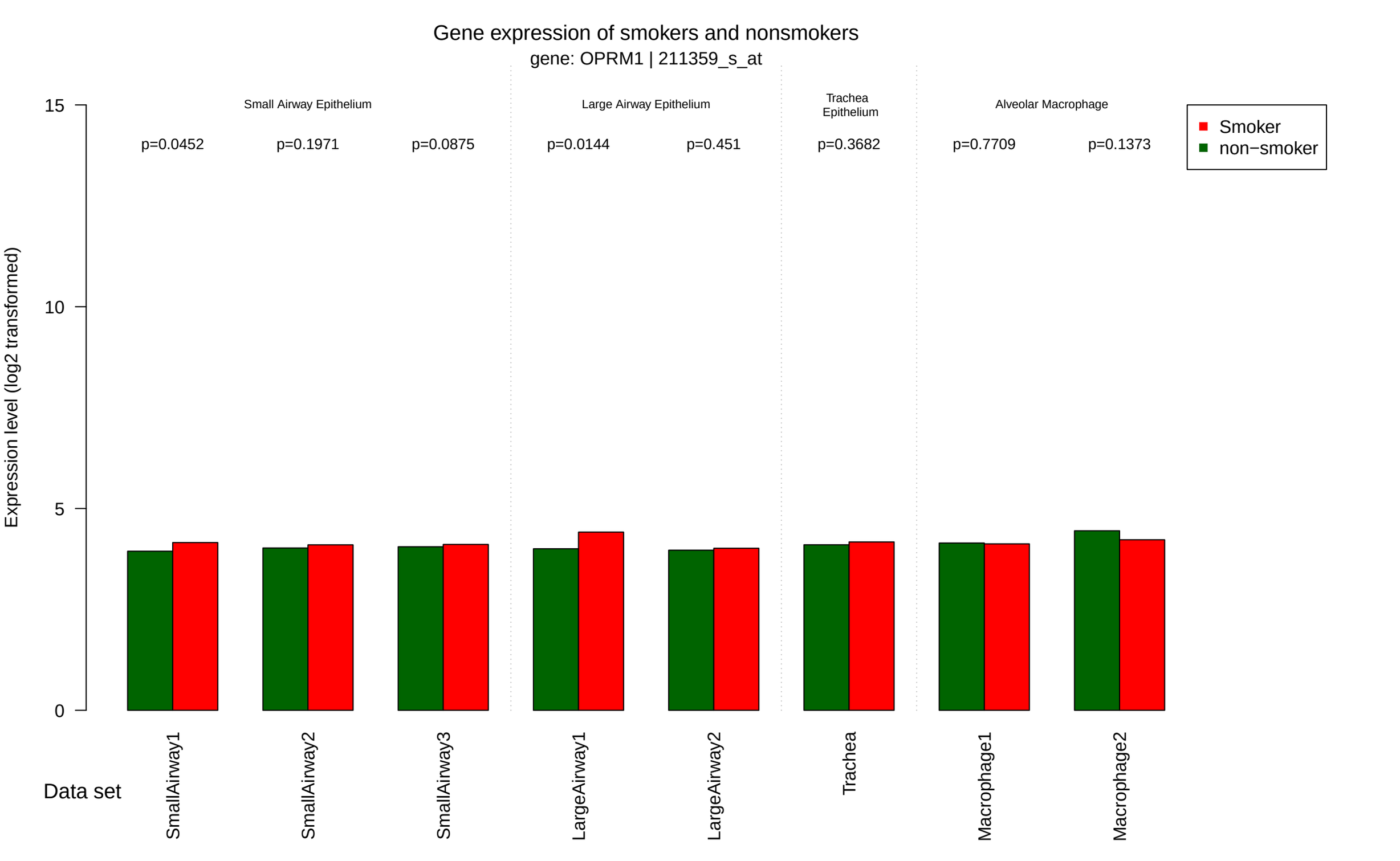
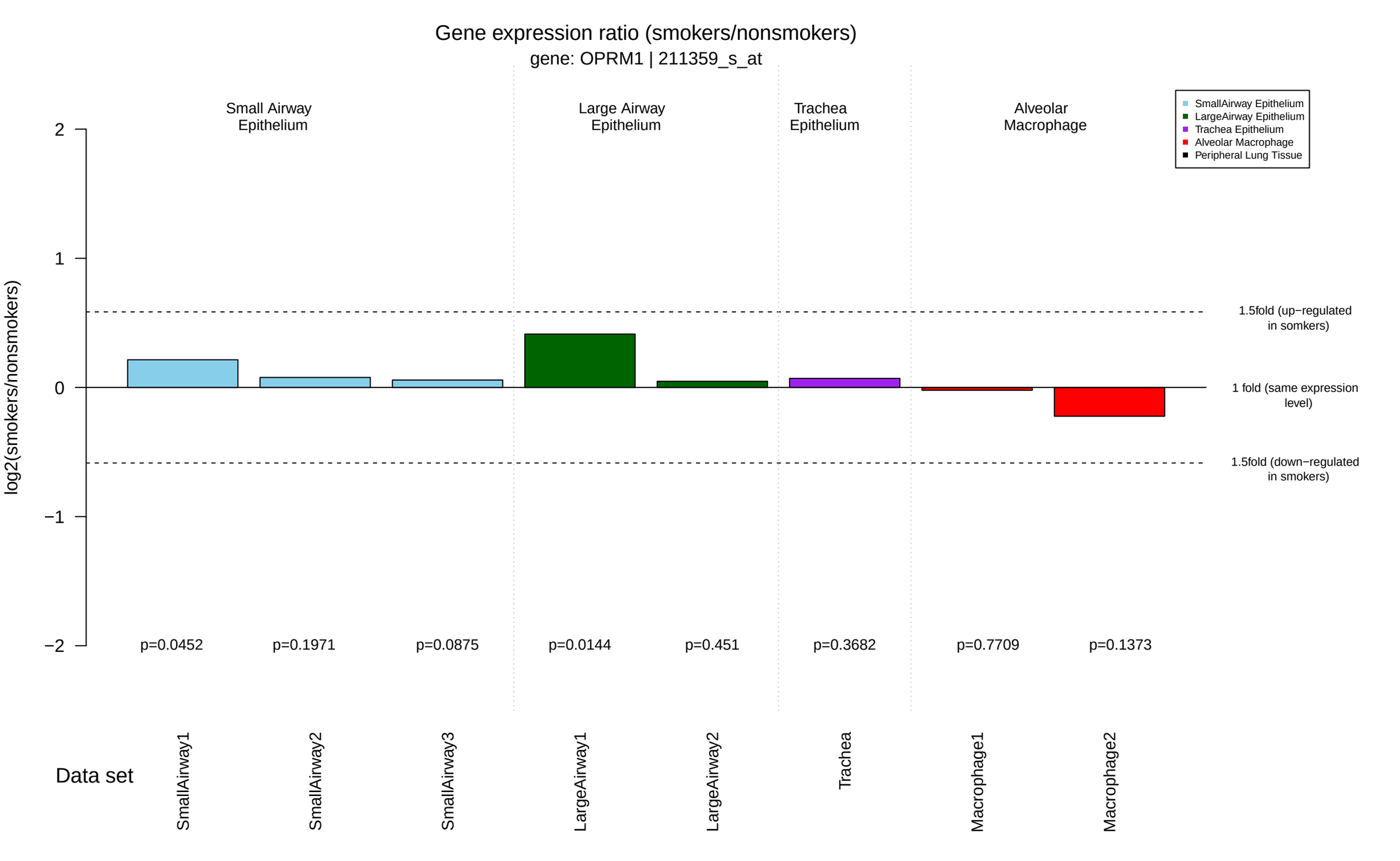
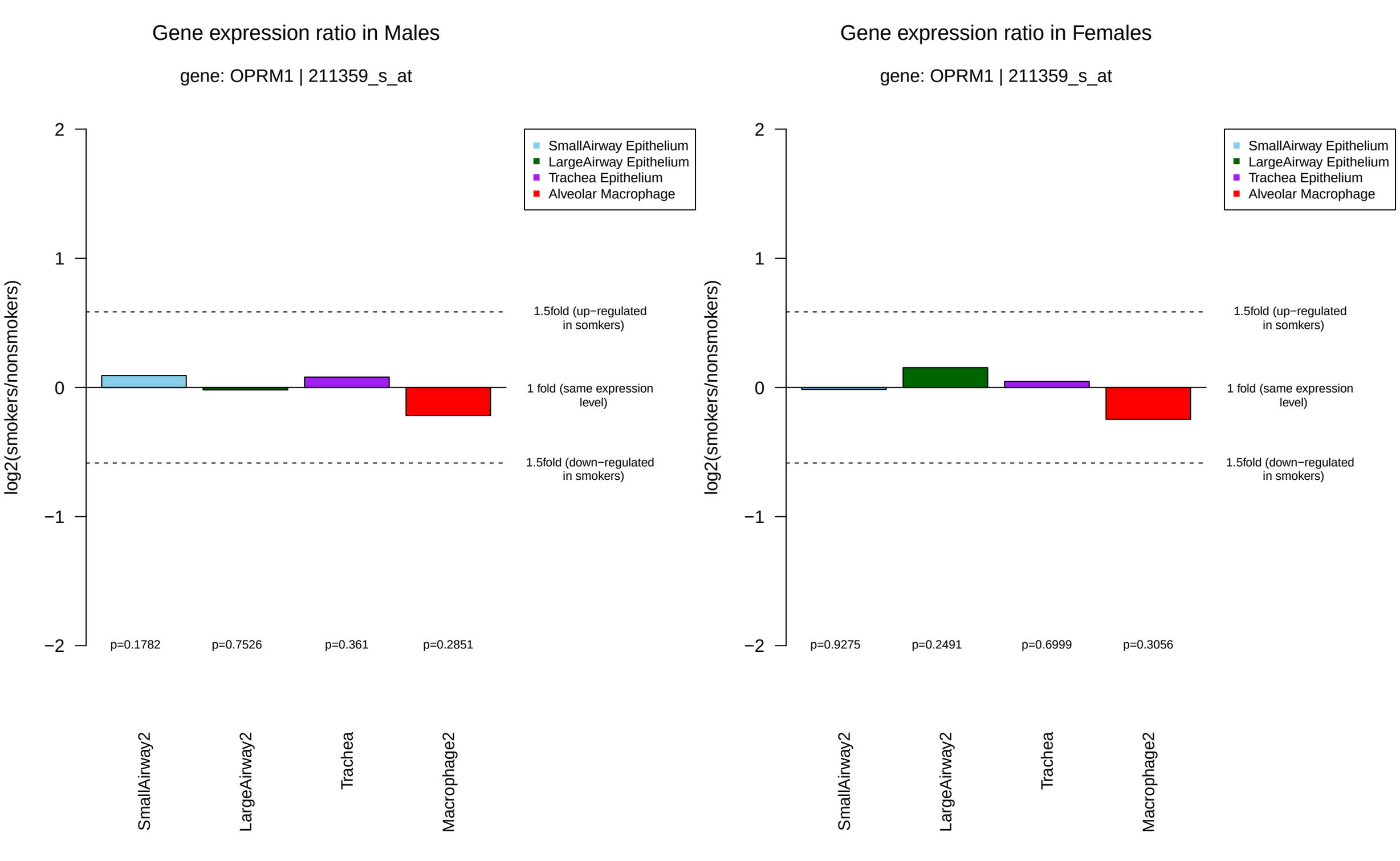
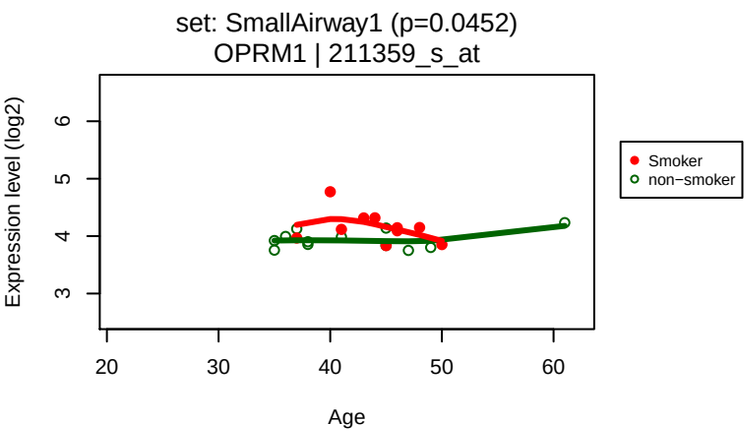
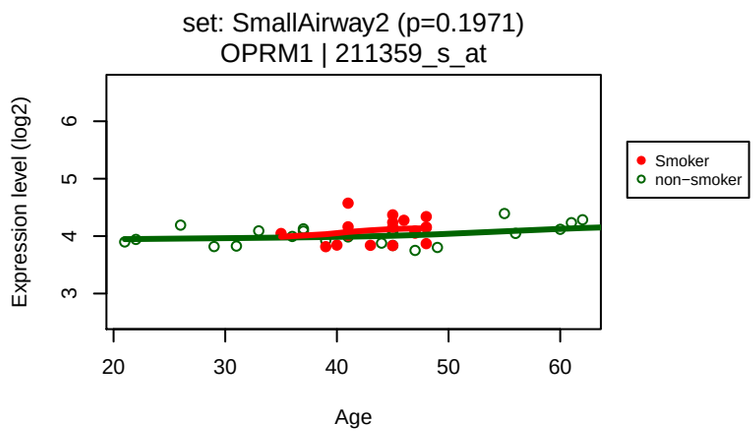
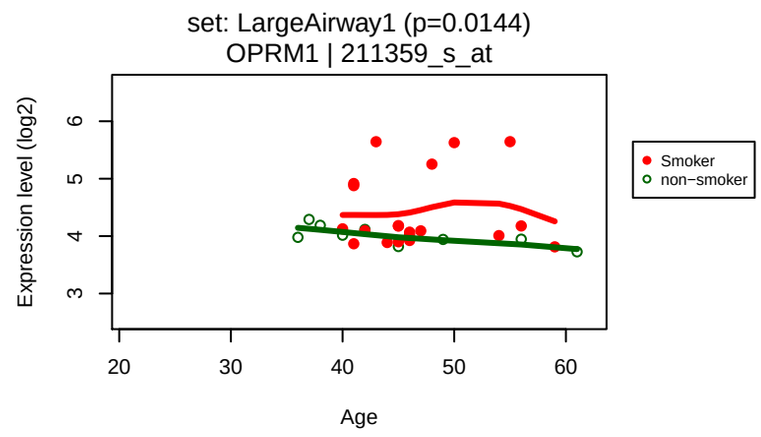

Gene expression of smokers and nonsmokers
gene: OPRM1 | 211359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0452
p=0.1971
p=0.0875
p=0.0144
p=0.451
p=0.3682
p=0.7709
p=0.1373
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

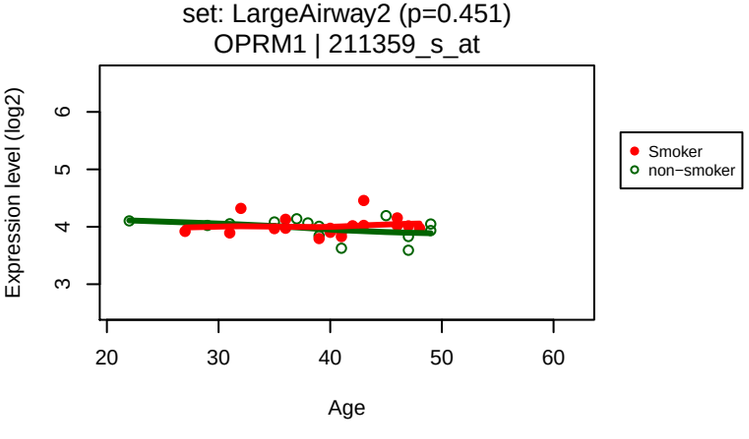

Gene expression ratio (smokers/nonsmokers)
gene: OPRM1 | 211359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0452
p=0.1971
p=0.0875
p=0.0144
p=0.451
p=0.3682
p=0.7709
p=0.1373
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

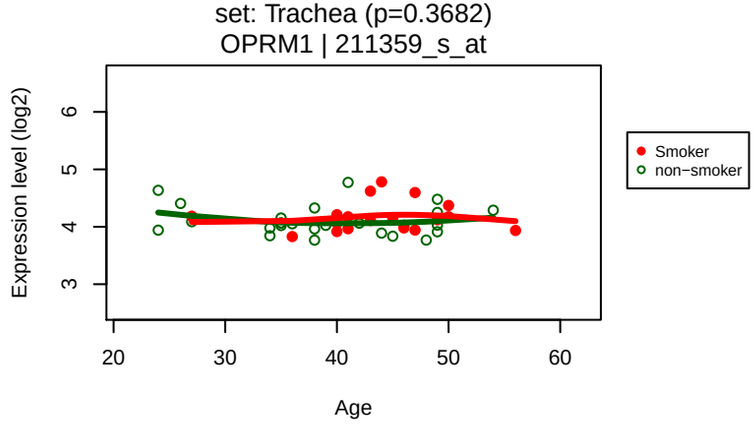

Gene expression ratio in Males
gene: OPRM1 | 211359_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1782
p=0.7526
p=0.361
p=0.2851
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OPRM1 | 211359_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9275
p=0.2491
p=0.6999
p=0.3056
SmallAirway2
LargeAirway2
Trachea
Macrophage2

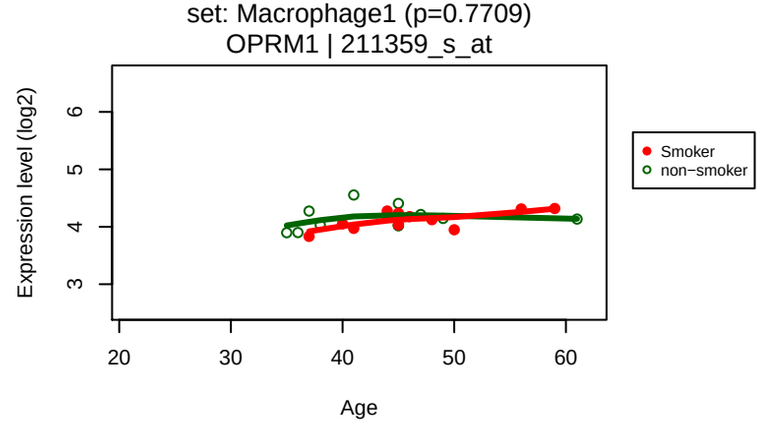

set: SmallAirway1 (p=0.0452)
OPRM1 | 211359_s_at
Expression level (log2)
Age
Smoker
non−smoker

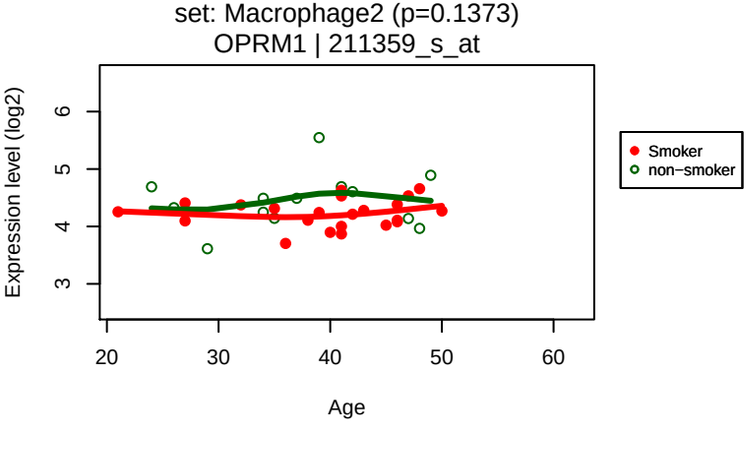

set: SmallAirway2 (p=0.1971)
OPRM1 | 211359_s_at
Expression level (log2)
Age
Smoker
non−smoker

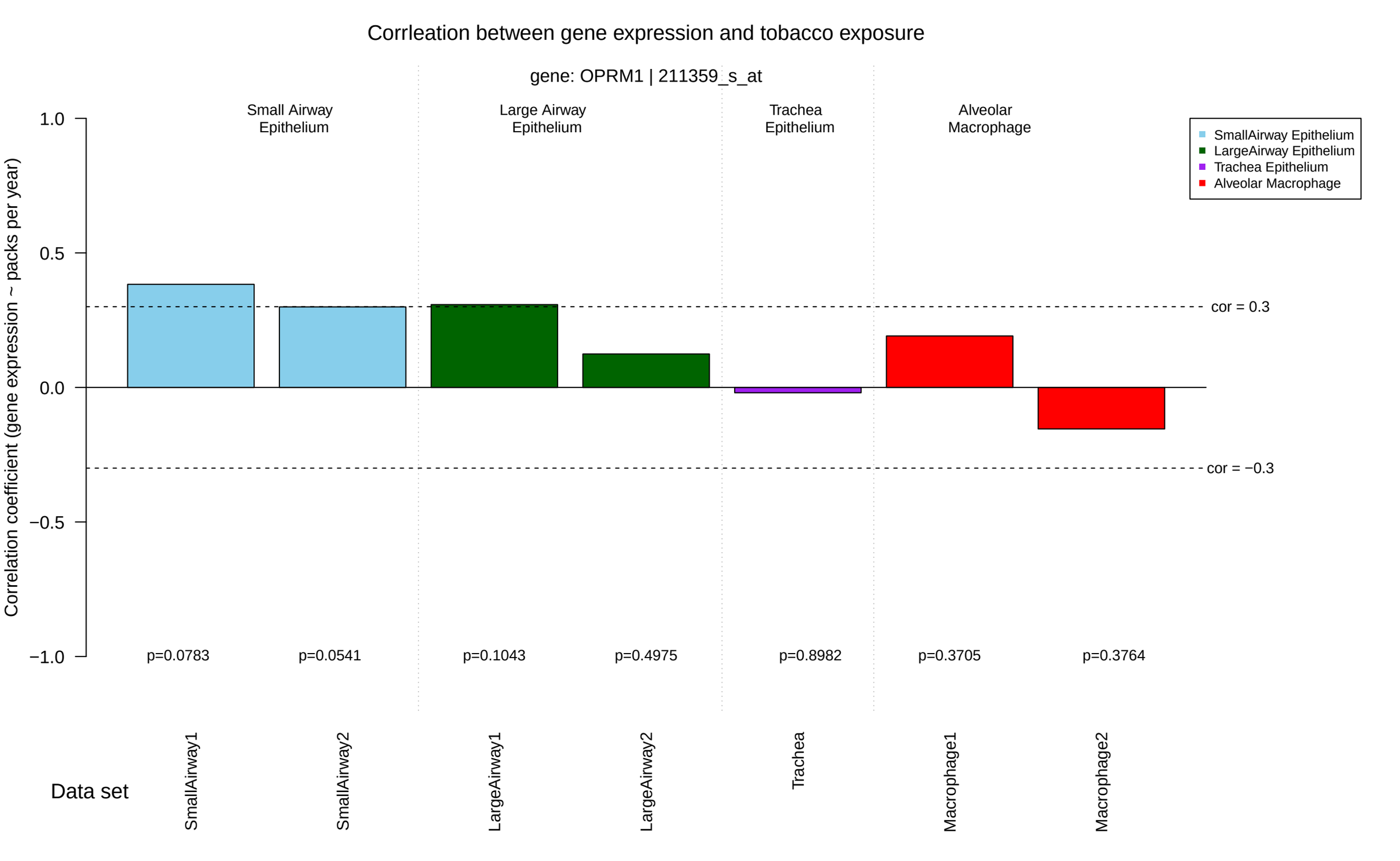

set: LargeAirway1 (p=0.0144)
OPRM1 | 211359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.451)
OPRM1 | 211359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3682)
OPRM1 | 211359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7709)
OPRM1 | 211359_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1373)
OPRM1 | 211359_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OPRM1 | 211359_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0783
p=0.0541
p=0.1043
p=0.4975
p=0.8982
p=0.3705
p=0.3764
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2